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Started on	2023-12-11 07:02
State	Finished
Completed on	2023-12-11 07:39
Time taken	36 mins 45 secs
Marks	5.00/5.00
Grade	**3.00** out of 3.00 (**100**%)

Question **1**
Correct
Mark 1.00 out of 1.00

Find true statements.

Select one or more:

- [x] Cross product of parallel vectors is zero vector. ✓
- [ ] Cross product of orthogonal vectors is zero vector.
- [x] Area of the parallelogram formed by two vectors is equal to the norm of their cross product. ✓
- [ ] Area of the triangle formed by two vectors is equal to the half of their dot product.

Your answer is correct.

The correct answers are: Area of the parallelogram formed by two vectors is equal to the norm of their cross product., Cross product of parallel vectors is zero vector.

Question **2**
Correct
Mark 1.00 out of 1.00

Find true statement(s).

Select one or more:

- [ ] If matrix is not invertible, then it is not diagonalizable.
- [x] If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable. ✓
- [x] If λ is an eigenvalue with multiplicity k, then corresponding eigenspace's dimension is k. ✗
- [ ] If an $n \times n$ matrix has less than n eigenvalues, then it is not diagonalizable.

Your answer is correct.

The correct answer is: If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable.

Question 3

Correct

Mark 1.00 out of 1.00

Let $\vec{u} = (2, 6, -7)$, $\vec{v} = (-1, -1, 8)$ and $k = 3$. If $k\vec{u} + l\vec{v} = (2, 14, 11)$, what is the value of l.

Select one:

- -4
- 2
- 4 ✓
- 0
- 1

Your answer is correct.

The correct answer is: 4

Question 4

Correct

Mark 1.00 out of 1.00

Find the area of the triangle with vertices $A(2, 3)$, $C(4, 7)$ and $D(-5, 8)$.

Select one:

- 18
- 15
- 10
- 12
- 19 ✓

Your answer is correct.

The correct answer is: 19

Question 5

Correct

Mark 1.00 out of 1.00

Let $\vec{p} = (2, k)$ and $\vec{q} = (3, 5)$. Find k such that $\vec{p}$ and $\vec{q}$ are parallel.

Select one:

- $\frac{5}{3}$
- $\frac{10}{3}$ ✓
- 1
- 2

Your answer is correct.

The correct answer is: $\frac{10}{3}$